Ipek Naz Obek

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Education

B.A., Cognitive Science University of Pennsylvania, Philadelphia, PA May 2025
Concentration: Computation and Cognition
Minors: Data Science, Computational Neuroscience
Summa Cum Laude; Distinction in Cognitive Science
Benjamin Franklin Scholar; Certificate in French
Honors Thesis: *Inferring Level Progress in Goal-Directed and Open-Ended Tasks*
Advisors: Dr. Alison Gopnik, Dr. Sharon Thompson-Schill, Annya Dahmani
Relevant Coursework: Programming Languages, Probability, Data Mining, Statistical Inference, Calculus I & II, Linear Algebra, Cognitive & Computational Neuroscience, Judgment & Decisions, Behavioral Economics, Developmental Psychology

Research Experience

Computation, Learning, and Development Lab Boston University, Boston, MA Aug 2025–present
Research Technician and Lab Manager
PI: Dr. Kate Nussenbaum

- Lead the design and implementation of online behavioral studies investigating goal pursuit strategies and persistence in children, adolescents, and adults.
- Design and program browser-based behavioral experiments using jsPsych, PsychoPy, and Qualtrics.
- Analyze trial-level behavioral data using mixed-effects models in R to characterize developmental and individual differences in learning and choice.
- Implement computational models of sequential decision-making in Python to test competing accounts of human exploration and value-guided learning.
- As the lab's first lab manager, established experimental, recruitment, and data-collection infrastructure, recruiting over 1,000 participants into the lab's database for developmental studies.
- Organize a biweekly multi-lab journal club bringing together researchers to discuss current papers.

Gopnik Cognitive Development and Learning Lab UC Berkeley, Berkeley, CA Jun 2024–May 2025
Research Assistant
PI: Dr. Alison Gopnik; Advisor: Annya Dahmani

- Designed a novel behavioral paradigm using Procgen environments to investigate how people infer learning progress in goal-directed and open-ended tasks.
- Developed computational accounts of learning-progress inference and compared human judgments with reinforcement-learning agents and large language models.
- Designed and analyzed experiments examining how people construct and evaluate curricula for complex, multi-step tasks.
- Supported by the Pinkel Research Fund, Benjamin Franklin Research Grant, and Career Services Research Grant; received the Benjamin Franklin Talk Prize.
- Recruited families on-site and conducted in-person behavioral testing with children in a museum setting, administering experimental protocols and collecting developmental data.

Thompson-Schill & Jenkins Lab University of Pennsylvania, Philadelphia, PA Jan 2024–May 2025
Research Assistant
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

- Conducted honors thesis research, jointly advised across the Thompson-Schill and Gopnik Labs, on human and AI representations of learning progress in goal-directed and open-ended environments.
- Participated in weekly joint lab meetings, discussing current research and literature with faculty and lab members.

Center for Neuroengineering and Therapeutics University of Pennsylvania, Philadelphia, PA Jun–Sep 2023
Research Assistant
PI: Dr. Brian Litt; Advisor: Dr. James Gugger

- Investigated relationships between traumatic-brain-injury lesion characteristics and subsequent post-traumatic epilepsy using multimodal neuroimaging data.
- Developed Python-based quantitative and machine-learning analyses relating lesion volume and location to clinical outcomes.
- Built an automated pipeline for MRI lesion segmentation and volumetric analysis, reducing reliance on manual annotation workflows.

Honors & Awards

Distinction in Cognitive Science University of Pennsylvania 2025
Awarded for completing an honors thesis, "Inferring Level Progress in Goal-Directed and Open-Ended Tasks," and for academic excellence in the major. Thesis advised by Alison Gopnik and Annya Dahmani (UC Berkeley Psychology) and Sharon Thompson-Schill (Penn Psychology).

Summa Cum Laude University of Pennsylvania 2025
Graduated with highest Latin honors from the College of Arts & Sciences.

Benjamin Franklin Talk Prize Benjamin Franklin Scholars, University of Pennsylvania 2025
Recognized for the best senior research presentation among the 2025 cohort of Benjamin Franklin Scholars, Penn's interdisciplinary honors program.

Pinkel Research Fund University of Pennsylvania 2025
Research funding awarded to support my senior honors thesis on curriculum learning in humans and machines.

Lucid Communication Challenge, Engineering and Physical Sciences Talk Penn National Research Conference 2024
Selected from a field of applicants at the National Research Conference at Penn to present my honors thesis research as a talk.

Benjamin Franklin Research Grant Benjamin Franklin Scholars, University of Pennsylvania 2024
Summer research funding awarded by the Benjamin Franklin Scholars Program; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

Career Services Research Grant Penn Career Services 2024
Competitive summer funding awarded by Penn Career Services; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

MindCORE Fellowship MindCORE, University of Pennsylvania 2023
Selected for a fully funded 10-week summer research program in cognitive science and neuroscience at Penn's hub for the integrative study of the mind.

Undergraduate Research Mentoring Award — Penn Undergraduate Research Mentoring Program — 2023

Fellowship awarded through Penn's Undergraduate Research Mentoring Program, which funds first- and second-year students to conduct summer research with a faculty mentor.

Dean's List — University of Pennsylvania — 2021–2025

Awarded in every academic year for a GPA of 3.7 or higher.

Benjamin Franklin Scholar — University of Pennsylvania — 2021–2025

Member of Penn's interdisciplinary honors program in the liberal arts and sciences.

Manuscripts in Preparation

Obek, I. N., Fu, J., Holton, E., & Nussenbaum, K. (2026). *Examining the development of goal commitment*. In preparation.

Presentations

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (2026, September). *Examining the development of goal commitment* [Poster]. Flux Society for Developmental Cognitive Neuroscience, La Jolla, CA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Poster]. Curiosity, Information Seeking, and Exploration Conference, Brown University, Providence, RI.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Talk]. Big Team Science Conference.
- **Obek, I. N.** (2025, April). *Machines that think like kids* [Talk]. Benjamin Franklin Scholars, University of Pennsylvania, Philadelphia, PA. Awarded the Benjamin Franklin Talk Prize.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Poster]. Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, August). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. MindCORE Research Fellowship Conference, Philadelphia, PA.

Teaching

Neuroscience and Society — Teaching Assistant, Penn Undergraduate Program in Neuroscience — 2024–2025

Instructor: Dr. Sharon Thompson-Schill

Supported students through weekly office hours, in-class instruction, and group activities, and provided individualized academic guidance.

Introduction to Cognitive Science — Teaching Assistant, Penn Undergraduate Program in Cognitive Science — Fall 2023

Instructor: Dr. Russell Richie

Led weekly recitations and office hours for 20+ students, developed instructional materials, and guided students through course concepts and assignments.

MATH 114: Calculus II — Tutor, Penn Weingarten Tutoring Center — Fall 2022

Tutored first-year students for 4 hours per week, providing individualized support with mathematical problem-solving and study strategies.

Mentorship

Benjamin Franklin Scholars Mentor Program — Peer Mentor, University of Pennsylvania — 2024–2025

Mentored 7 Benjamin Franklin Scholars in their academic and social transition to Penn through regular individual outreach and community-building events.

Center for Undergraduate Research and Fellowships — Research Peer Advisor, University of Pennsylvania — 2024–2025

Advised undergraduates pursuing research opportunities and organized research programming, including an international faculty symposium on the experiences and challenges of international academics.

Pioneer Academics — Program Coordinator and Advisory Board Member — 2022–2025

Advised 100+ students through the research and writing process and coordinated research seminars on effective research practices and faculty mentorship.

Young Guru Academy — Alumni Mentor — 2019–present

Mentor Turkish high school and university students pursuing international education and STEM careers. Led STEM outreach across 5+ cities in rural Anatolia, training teachers and students and delivering hands-on science education to underserved communities; presented at YGA's annual summit to ~20,000 attendees.

Leadership & Service

Cognitive Science Student Association — Founder & President, University of Pennsylvania — 2024–2025

Founded and led Penn's Cognitive Science Student Association, organizing faculty talks, research seminars, mentoring, and networking initiatives connecting students with faculty, alumni, and industry professionals.

The DREAM Program — Recruitment Chair, Philadelphia, PA — 2023–2025

Led volunteer recruitment and weekly educational programming for preschool-aged children in West Philadelphia, including hands-on science activities designed to foster curiosity, collaboration, and confidence in learning.

Turkish Student Association — President, University of Pennsylvania — 2022–2025

Directed a 50-member organization and community initiatives, including raising $8K for earthquake-relief efforts.

Assembly of International Students — Social Committee Member, University of Pennsylvania — 2021–2025

Organized social and cultural programming to foster community among Penn's international student population.

The Daily Pennsylvanian — Photographer, University of Pennsylvania — 2021–2025

Photographed campus events and contributed visual content for multimedia stories at Penn's student newspaper.

Skills

Programming: Python, R, JavaScript, MATLAB, SQL, Java, OCaml

Research tools & platforms: jsPsych, PsychoPy, Qualtrics, Prolific, Pavlovia, Git/GitHub, Jupyter, LaTeX, AFNI

Certifications: CITI Training (Biomedical Responsible Conduct of Research; Human Subjects Research; Social-Behavioral Research with Children)

Languages: Turkish (native), English (fluent), French (fluent; Certificate in French, University of Pennsylvania)

References

Dr. Kate Nussenbaum
Assistant Professor, Department of Psychological and Brain Sciences
Boston University
katenuss@bu.edu

Dr. Alison Gopnik
Professor of Psychology
University of California, Berkeley
gopnik@berkeley.edu

Dr. Sharon Thompson-Schill
Christopher H. Browne Distinguished Professor of Psychology; Co-Director, Center for Cognitive Neuroscience
University of Pennsylvania
sschill@psych.upenn.edu

Dr. Russell Richie
Director of Strategy and Impact
Progress and Poverty Institute
Formerly Associate Director, Cognitive Science and MindCORE, University of Pennsylvania
drussellmrichie@gmail.com

Annya Dahmani
PhD Student, Department of Psychology (advised by Alison Gopnik)
University of California, Berkeley
adahmani@berkeley.edu